Deauth n' Crack

An Introduction to Wireless Attacks: WPA2, Deauth, Capture, and Crack

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Another one got caught today, it's all over the papers. "Teenager Arrested in Computer Crime Scandal", "Hacker Arrested after Bank Tampering"...

Phrack issue #40 — The Mentor

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Attempting to follow the recommendations or implement the ideas in this paper may not fit your context, may not solve your problem, and may in certain circumstances plunge your program and career into abject chaos.

Figure 1: Do not try this at home, kids[4]

1 What

In this write-up, I will discuss how one may approach cracking a Wireless Access Point's (WAP) password.

This can be done in a couple of steps:

1. Find your Access Point BSSID[1]
2. Monitor the BSSID for connections
3. Force devices to reconnect to the WAP through forced deauthentication (optional)
4. Convert the captured WPA2 Handshake[2] into a crackable format
5. Crack the file

It is not necessary to force devices to reconnect (Deauthentication Attack[3]), but it does speed up the capture process.

2 How

For this demonstration, a specific set of tools will be used, as well as any external wireless card, as long as it can run in monitor mode. Specifically `aircrack-ng`, `hcxtools`, and `Hashcat` (or even `John the Ripper`); all of which we will use on a Linux based OS (In this case x86-64 Debian).

Hcxtools provides a program named `hcxpcapngtool`, which converts the captured WPA2 Handshake (which will be discussed latter on) into a format we can use to crack.

[1] A BSSID (or Basic Service Set Identifier) is a 48-bit label that conforms to the MAC-48 conventions. A device can have multiple BSSIDs, which are similar to MAC addresses.[1]

This *label* is used with a service set. A service set is a group of Wireless Networks (not exclusively WAPs) which all share a SSID (Service Set Identifier). A SSID is a human readable identifier; they are used in broadcasts as beacon packets to announce itself.[1]

[2] Wi-Fi Protected Access (WPA) was introduced to fix the "completely broken" Wired Equivalent Privacy (WEP). There are multiple version of WPA (WPA1, WPA2, WPA3), each improving on the previous one. This write-up will be attacking WPA2.[5][6]

[3] A deauthentication attack is considered a type of denial-of-service (DoS) attack.[2] The key differences between a normal DoS and a *deauthentication* attack is in how they work. A deauth attack sends "disassociate packets to one or more clients," which forces the targeted clients to disassociate with the network, forcing them to re-establish the connection.[3]

For our purposes, this does not have to be a flood of packets; it can be just one.

2.1 Setup

For a simple setup, you will need to accomplish a couple things:

1. Get an external wireless adapter[4]
2. Install necessary packages
3. Stop any programs that are using the wireless interface
4. Enable Monitor mode on the specified wireless interface[5]

An example of this is as follows:

```
# After you plug in the adapter (1), and get it working,
# you can (2) install these packages:
sudo apt install aircrack-ng hcxtools hashcat

# To stop any interfering programs (3),
# you need to know which programs are interfering.
sudo airmon-ng check
# Appending 'kill' will remove the offending processes.
sudo airmong-ng check kill

# There are multiple ways to enable monitor mode (4).
# First you will need to find your interface.
ip addr show
# Then Airmon-ng provides an easy way.
sudo airmon-ng start <interface>
```

2.2 Find

Next we will find the BSSID and channel of our target WAP. There is a program installed with `aircrack-ng` called `airodump-ng`. This is a simple wireless packet capture tool (there are many others). If we use this tool, with our designated interface as its only argument, it will show all wireless networks in the vicinity along with their BSSIDs, SSIDs and Channels used.

An example of its use is as follows:

```
sudo airodump-ng <interface>
```

Once you have found your specified target SSID make sure to note down the BSSID and Channel used.

[4] You could technically use your host wireless network adapter. This does, however, bring some difficulties, for example, monitor mode not being supported, or inability to connect to the internet during attack.

[5] To be able to intercept traffic monitor mode (or equivalent)[7] needs to be enabled on the NIC. This allows the NIC to receive and monitor all traffic on a channel.

2.3 Watch

After the BSSID and Channel number is found, we can start targeting. To ensure we capture the handshake, we will have to narrow our search. With a similar command, you can use the bssid and the channel number to narrow where to look.

The command could look something like the following:

```
# `--bssid`: specifies the BSSID to search for.
# `--channel`: specifies the Channel to search on.
# `-w`: specifies the dump file prefix name.
sudo airodump-ng <interface> --bssid <bssid> --channel <channel> -w
    <output>
```

If you receive a handshake, you should see a *WPA Handshake* message after the date and time section on the `airodump` output. It should look something like the following:

```
CH  1 ][ Elapsed: 10 s ][ 2023-10-27 10:32 ][ WPA handshake: ...
BSSID            PWR  RXQ  Beacons   #Data, #/s CH  MB   ENC  CIPHER AUTH ESSID

00:11:22:33:44:55 -32 100      25       10   1   1  54e  WPA2 CCMP  PSK
    LabEx_WiFi

BSSID            STATION          PWR   Rate    Lost    Frames  Probe

00:11:22:33:44:55 AA:BB:CC:DD:EE:FF -40  1- 1     0       15
```

2.4 Deauthentication (optional)

Next the actual "attack" on the WAP. Do note: it is not necessary to do a deauthentication attack, you could wait paitently and wait for a new connection or reconnection from a client.

We will send Deauthentication packets to our target. You will need to know a host *MAC Address*, since we are deauthenticating between the WAP and a Client. The tool `aireplay-ng` will allow us to do this.[6]

An example of this is as follows:

```
# `--deauth`: Specifies deauthentication attack,
#  and the amount of times to send. Zero for infinite.
# `-a`: Specifies the Access Point to Attacks BSSID
# `-c`: Specifies the Clients MAC Address to Attack
sudo aireplay-ng --deauth <num> -a <ap> -c <client> <interface>
```

2.5 Convert

Since the handshake is captured, to proceed, you will need to extract the handshake in a way that we can use in hashcat (used in this example). The file we will be using is the PCAP file (the one ending in *cap*).[B] With this capture file, however, not much can be done – when it comes to this write-up – without extracting the handshake.[7]

So to extract the handshake we will need to use the second tool we needed: *hcxtools*.[8] With the `hcxpcapngtool` that comes with it, we can convert raw captures to hashcat readable formats.[12] That can be done with the following command:

```
hcxpcapngtool -o <output_file> <input_file>.cap
```

[6] The manual page for `aireplay-ng` says the following in reference to the deauthentication attack mode:

> This attack sends deauthentication packets to one or more clients which are currently associated with a particular Access Point. Deauthenticating clients can be done for a number of reasons: Recovering a hidden ESSID. This is an ESSID which is not begin broadcast. Another term for this is "cloaked" or Capturing WPA/WPA2 handshakes by forcing clients to reauthenticate or Generate ARP requests (Windows clients sometimes flush their ARP cache when disconnected). Of course, this attack is totally useless if there are no associated wireless client or on fake authentications.[9]

[7] You could load this into a network analyser, for example `Wireshark`, or any other program using `libpcap` or equivalents.

[8] You can find most distobutions packages on `https://pkgs.org/` under the hcxtools package.

2.6 Crack

Once you have the correct file, we can now move on to the cracking, which is heavily dependent on your setup and target. That subject on its own could be an entirely seperate write-up. The basic command for cracking the file is as follows:

```
# ‘-m‘: The mode, or hash type.
# ‘-a‘: Attack mode, (0 == dictionary).
# <hash_file>: The convert output_file from the PCAP file
# <wordlist>: The wordlist of potential passwords
hashcat -m 22000 -a 0 <hash_file> <wordlist>

# After the previous command has finished,
# use the ‘--show‘ flag to show the pot file,
# or the file with the cracked password.
hashcat -m 22000 -a 0 <hash_file> <wordlist> --show
```

This method of crackind is heavily dependent on the wordlist, which may or may not have the password for the AP in it. There are many ways to guess inteligently, using rules, or custom dictionaries; but again thats out of scope.

3 Remediation

There is not really any way to “remediate” this flaw in WPA2, it is a fundamental issue with the WPA2 protocol. You could increase the passkey to a resonable amount of entropy, but that would only patch the problem. It turns out there are other vulnerabilities with WPA2.[31][Table 4]

The best way to prevent this is to not have one (best part is none). But if you have to, reduce who can access it – meaning decrease the range to only inside the building – and use only WPA3-Enterprise with a good password (or none and use 802.1X[29]), and good employee training.[9]

4 Conclusion

For WPA2 PSK wireless networks, it is deceptively simple to get your hands on the hadshake between clients. With aproxamently 75% of networks use WPA2, the only problem is comming from the actually cracking the handshake.

Even if you block this path, however, there are also other ways of getting into a network, for example, WPS bruteforcing. But you do not have to limit yourself to the electronic realm, you could just break in, or talk to the right person at the right time (along with a clip board).

[9] This is not a complete list, or a completly sound recommendation. see: [Figure 1]

A Wi-Fi

A.1 Standards

There are multiple standards, and groups of standards. There is the *IEEE 802.11* (Institute of Electrical and Electronics Engineers) standards and some wireless security protocols – WPA (Wi-Fi Protected Access) and the older WEP (Wired Equivalent Privacy) protocols – which we will focus on.

A.1.1 IEEE 802.11

For specifying wireless computer communication protocols (or Wi-Fi), the IEEE 802.11 protocol – a part of the IEEE 802 (a standard for defining area networks) – which comes in multiple subsections. These subsections are denoted by letters appended to the 802.11 standard number. In 2018, however, the Wi-Fi Alliance[10] started to use generations for naming the publicly used 802.11 protocols.[13][14][15][Table 1]

Gen.	**IEEE standard**	**Adopt.**	**Link rate (Mbit/s)**	**RF (GHz)**		
				2.4	**5**	**6**
Wi-Fi 1	802.11	1997	1–2	X		
Wi-Fi 2	802.11b	1999	1–11	X		
Wi-Fi 2G	802.11a		6–54		X	
Wi-Fi 3	802.11g	2003		X		
Wi-Fi 4	802.11n	2009	6.5–600	X	X	
Wi-Fi 5	802.11ac	2013	6.5–6,933		X	
Wi-Fi 6	802.11ax	2021	0.4–9,608	X	X	
Wi-Fi 6E				X	X	X
Wi-Fi 7	802.11be	2024	0.4–23,059	X	X	X

Table 1: 802.11 Generations[17]

[10] The *Wi-Fi* Alliance is a organization that ensures the interoperability between equipment throught their certification program and their ownership of the Wi-Fi trademarks.[16]

A.1.2 WPA2 4-Way Handshake

The 4-*Way Handshake*, the object we want to capture, is an authentication process (using the IEEE 802.1X process[29]). This process is carried out over four messages (in other words: 4-Way Handshake), between the authenticator (AP) and the supplicant (STA).[27][28, 8.5.3][30][Figure A.1.2]

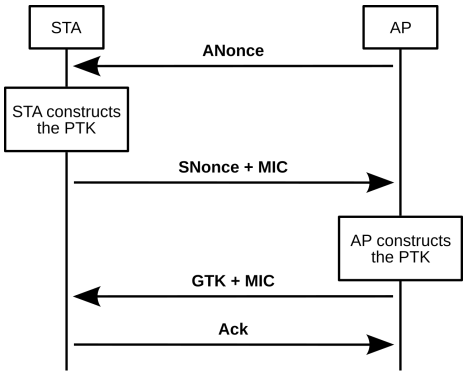

Figure 2: Four way handshake[27]
STA (Station/Client) authentication handshake with AP (Access Point)

ANonce or message 1, is the AP sending an EAPOL message with Anonce (a random number) which it uses to make a key (to encrypt traffic). **SNonce+MIC** or message 2, sends a *SNonce*, so the AP can make a key as well, with a MIC (Message Integrity Check) to ensure the *SNonce* was not corrupted. **GTK+MIC** or message 3, sends the GTK (Global Temporal Key), which allows all broadcast and multicast traffic to be encrypted, along with another MIC. **Ack** or message 4, just confirms that the keys have been installed, or Acknowledge.[30]

	Adopt.[11]	**Enc.**	**Key Size**
WEP	1997	RC4	64-bit & 128-bit
WPA	1999[12]	TKIP+RC4	128-bit
WPA2	2004	AES-CCMP	128-bit
WPA3	2018	AES-CCMP/AES-GCMP[13]	128-bit & 256-bit[13]

Table 2: Wireless Security Protocols[17][18]

A.1.3 Security Protocols

There has been two sets of security protocols for wireless area networks, WEP (Wired Equivalent Privacy) and WPA (Wi-Fi Protected Access). The Wi-Fi Aliance[10] defined WPA in response to the disastrous failure of WEP, it should not be used in any respect.

There are multiple known security issues with WPA, it would be recommended to pick the latest version of WPA.[Table 4]

There is also another vector of potential attack, WPS (Wi-Fi Protected Setup). WPS allows devices to connect to a small office and home office (SOHO) networks without the input of a long passphrase. WPS can connect without any password, or using WPS PIN.

A major security flaw was revealed (Dec. 2011), in WPS PIN, which allows an attacker to recover a WPS PIN in 4-10 hours using brute-force.

[11] This column is for when each protocol was first introduced. This does not include the date when they were required.

[12] "WPA could be implemented through firmware upgrades on wireless network interface cards designed for WEP that began shipping as far back as 1999. However, since the changes required in the wireless Access Points (APs) were more extensive than those needed on the network cards, most pre-2003 APs were not upgradable by vendor-provided methods to support WPA."[17]

[13] WPA3 uses two different modes, WPA3-Enterprise mode, and WPA3-Personal mode. Enterprise uses AES-256 in GCM mode. Personal uses CCMP-128 (AES-128 in CCM mode).[17]

[14] Forward Secrecy ensures that data from a seperate session will not be comprimised, limiting the damage in case the secret is comprimised. With WPA and WPA2, if you know the secret, you can "potentially" decrypt all packets.[19]

[15] Hole196 abuses the shared Group Temporal Key (GTK), using it to conduct man-in-the-middle (MitM) and denial-of-service attacks.[20] This assumes, however, that the attacker is already authenticated.

Vuln.	WPA	WPA2	WPA3
Password Cracking	X	X	
Forward Secrecy[14]	X	X	
Hole192[15]		X	
Predictable Group Temporal Key	X	X	
WPS Brute Force[21]	X	X	
KRACK Attack		X	
Dragonblood			X
FragAttack	X	X	X

Table 3: WPA Known Security Issues[17]
Some of the know security issues directly caused to WPA.

A.2 Channels

Channels, in this context, are frequencies accessed while using *IEEE 802.11* protocols. The most common frequency ranges, in which the channels are selected from, are: 2.4 GHz, 5 GHz, and 6 GHz depending on the 802.11 protocol in use. Different channels are used to prevent collison and interference from other nearby WAPs. [22][A.1.1][Table 1]

B PCAP files

Pcaps (or Packet Captures), defined in the libpcap library, are used to capture network trafic and store it as a file. The term pcap generally refers to the API, often as the *libpcap* library (for "Unix-like systems").[16]

There are two types of PCAP files now, since the first one is being phased out: Pcap,[17] and PcapNG (Packet Capture New Generation). Both of these formats, however, are designed to store packets. The main difference in PcapNG is it allows for the storage of packets with different link layer types.[23][24][25][18]

A more thorough list of differences between Pcap and PcapNG are as follows:

- "Multiple PcapNG capture files can be merged into one simply by concatenating them"[23]
- "Allows comments, also known as 'annotations', to be attached to packets"[23]
- "Data structures designed specifically for storing metadata, such as hostnames, encryption keys, capture filters and information about the interface used to capture the packets"[23]

[16]Windows does have ports: *WinPcap*, no longer supported/developed, and *Npcap*.
[17]Pcap files are also refered to as libpcap files.[23]
[18]The link layer is lowest layer in the TCP/IP stack, and defines how your link, or your physical network card, talks to adjacent network nodes (for example a router).

C WPA Statistics

Encryption	Device Numbers	(%) Total	(%) Known[19]
WPA3	71,030,652	3.86%	4.64%
WPA2	1,372,980,560	74.62%	89.70%
WPA	39,325,719	2.14%	2.56%
WEP	47,165,331	2.56%	3.08%
????	277,379,384	15.08%	
None	32,741,727	15.08%	

Table 4: WPA Security Statistics[32][33]
These are general statistics about Wireless Encryption from WiGLE[32]

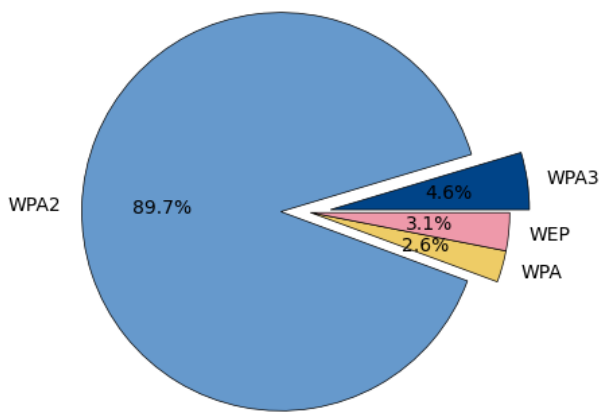

Figure 3: WPA Security Statistics Pie Chart[32][33][34]
General statistic about Wireless Encryption[19]

[19] The numbers in this column was calculated by taking the total number of devices that are known to use WPA3, WPA2, WPA1 or WEP. The equation for each line is as follows

$$percentage_x = \frac{a_x}{\sum_{i=1}^{n} a_i}$$

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